Ziji Yu

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Research Interests: Trustworthy Multimodal Learning; Vision-Language Models; Retrieval-Augmented Generation; Human-AI Collaboration

EDUCATION

Stony Brook University, College of Engineering and Applied Science — Stony Brook, NY
B.S. in Information Systems; Double Major in Economics — Jan 2023 - Expected Dec 2026
GPA: 3.4/4.0 | GRE: 335 (V 165, Q 170)
Core Course: Computer System (A), Mathematical Statistics (A), Data Science & Machine Learning (A), Introduction of Networking (A)

RESEARCH EXPERIENCE

Zhejiang University, State Key Laboratory of CAD&CG — Hangzhou, China
Research Assistant | Advisor: Prof. Wei Chen — Mar 2025 - Jul 2026

Research program : *Trustworthy and retrieval-grounded multimodal learning for cultural heritage.*

- **Zhihua - Domain-Specialized Vision-Language System:** Curated and mined a multi-source corpus of 100,000+ traditional Chinese paintings; developed image-QA generation and refinement pipelines; constructed preference pairs for DPO; fine-tuned and evaluated a Qwen2.5-VL model variant; and implemented the retrieval-augmented generation pipeline.
- The resulting system achieved the top overall benchmark score (65.8 vs. 57.7), ranked first in seal recognition and hallucination detection, and received user ratings of 4.35/5 for accuracy and 4.19/5 for satisfaction (n=75). **Hugging Face:** zjuqx/ZhiHua **Website:** https://zhihua.zjuqx.cn/
- **Local-to-Global Visual Retrieval and Grounding:** Developed the two-stage patch-level vector retrieval and LightGlue reranking system; built the top-k inspection interface; and integrated local- and artwork-level evidence into painting Q&A.
- The system maps partial painting crops to source artworks and regions, providing inspectable evidence for retrieval-grounded responses.
- **SILK - Sparse Knowledge and OOD Technique Recognition:** Built an expert-validated ID/OOD benchmark containing 1,014 images and 3,042 annotations across 12 seen and 11 held-out painting techniques.
- Contributed to continued pre-training, full-parameter SFT/RLVR, and inference-time reference augmentation.
- **ArtFlow - Training-Free Agentic Painting Analysis:** Implemented components of the retrieval-grounded perception, parallel reasoning, validation, and reflection workflow, and contributed to an expert-informed evaluation covering 1,560 records from 199 paintings.
- The workflow improved GPT-4.1 dimension coverage by 14.1% and exploration-aware coverage by 63.6%; 73.1% of samples covered all 15 appraisal dimensions. **GitHub:** ZijiYu/ArtFlow
- **Domain-Bootstrapped OCR - Low-Resource Inscription Recognition:** Built the five-stage adaptation pipeline spanning multi-model candidate generation, expert verification, filtered pseudo-labeling, and retraining; created a 942-sample seed set and retained 4,943 pseudo-labels.
- The adapted model reached F1 scores of 0.853 on validation and 0.840 on the seed benchmark versus 0.810/0.762 for the base model; ablations confirmed gains from data scale.

MANUSCRIPTS

Zhihua: Leveraging Vision Language Models for Traditional Chinese Painting Appreciation and Storytelling
W. Chu, X. Deng, L. Ding, ..., **Z. Yu**, ..., W. Chen | **Revise and resubmit at npj Heritage Science**

SILK: A Sparse Domain Knowledge Learning Framework for Traditional Chinese Painting Technique Recognition
L. Ding, J. Ke, J. Lu, ..., **Z. Yu**, ..., W. Chen | **Submitted to AAAI 2027**

ArtFlow: An Agentic Workflow for Retrieval-Grounded Traditional Chinese Painting Analysis
L. Jiang, **Z. Yu**, B. Zhu, ..., W. Chen | **Submitted to AAAI 2027**

Domain-Bootstrapped OCR for Low-Resource Traditional Chinese Painting Inscriptions
W. Chu, ..., **Z. Yu**, ..., J. Ke, X. Deng, W. Chen | **Submitted to ICLR 2027**

When Generalization Goes the Wrong Way: The Limits of Zero-Shot Offline Meta-RL
X. Zhang, **Z. Yu**, L. Li, K. Mueller | **Submitted to ICLR 2027**

INTELLECTUAL PROPERTY

Method and System for General-Purpose Multimodal Image Evidence-Enhanced Closed-Loop Analysis 2026

Patent application draft: Designed a domain-configurable, slot-based multimodal image analysis system that cross-validates visual and retrieved evidence through multi-perspective negative review, retrieval routing, and iterative state feedback; added human slot control, white-box trace logging, and convergence safeguards for auditable use across real-time video, medical imaging, industrial inspection, remote sensing, and art analysis.

TECHNICAL SKILLS

Models & Methods: Vision-language models post training, full-parameter SFT, DPO, RLVR, retrieval-augmented generation, visual grounding, pseudo-labeling, and out-of-distribution evaluation
Programming & Tools: Python, Java, SQL, R, PyTorch, ms-swift, Vllm, Docker, Git, Linux
Languages: Chinese (native), English (fluent)

ENGINEERING EXPERIENCE

Anker Innovations Co., Ltd. Shenzhen, China
Software Testing Intern May 2024 - Aug 2024

- Engineered a Docker-based production-parity simulation stack and optimized resource limits and connection pools, cutting error rate to under 5%; built Python BDD suites that increased critical-scenario coverage by 30% and reduced manual testing time by 50%.

ACADEMIC SERVICE, TEACHING & LEADERSHIP

WPES 2025 - ACM Workshop on Privacy in the Electronic Society Student Reviewer
Microeconomics Undergrad TA Feb 2023 - May 2023

Led weekly office hours and recitation sessions for 10+ students and maintained assignment and progress records.